

# Output Current Parameterization & Slope Sizing Flow

## 1. Testbench Configuration

**Testbench:** PrimeSim.default\_final

**Design:** Tese/Vtemp\_gen\_rod/schematic\_rod\_with\_opamp\_final

**Analyses:**

- **DC Temperature Sweep (dc):**  $-40^{\circ}\text{C}$  to  $125^{\circ}\text{C}$  (step  $16.5^{\circ}\text{C}$ )
- *Loop Stability (lsta):* Logarithmic sweep from 1 MHz to 13 GHz (Defined but disabled in flow)
- *Transient (tran):* 0 to 5  $\mu\text{s}$  with a 50 ns step (Defined but disabled in flow)

**Corners:** TT, SS, FF (Tandem Temperature Sweeps with respective Process Models)

## 2. Output Expressions

| OUTPUT                                      | EXPRESSION                                              |
|---------------------------------------------|---------------------------------------------------------|
| $V_d$                                       | v(/Vd)                                                  |
| $V_r$                                       | v(/Vr)                                                  |
| $I_{\text{temp}}$                           | i(/R29/PLUS)                                            |
| $I_{25}$ (Current at $25^{\circ}\text{C}$ ) | yvalue(Itemp, 25)                                       |
| $\Delta T$ (dT)                             | xmax(Itemp) - xmin(Itemp)                               |
| $\Delta I$ (di)                             | yvalue(Itemp, xmax(Itemp)) - yvalue(Itemp, xmin(Itemp)) |
| Slope ( $\frac{di}{dT}$ )                   | di / dT                                                 |
| Normalized Slope                            | di_over_dT / i25                                        |
| $V_{\text{outAmp}}$                         | v(/VoutAmp)                                             |
| $V_{\text{temp}}$                           | v(/Vtemp)                                               |
| Total Current                               | abs(i(/V9/PLUS)) - 90u                                  |
| $V_x$                                       | v(/I15/Vx)                                              |
| $V_{\text{itail}}$                          | v(/I15/net74)                                           |
| Gain Margin                                 | meas(gain_margin)                                       |
| Gain Margin Frequency                       | meas(gain_margin_freq)                                  |
| Phase Margin                                | meas(phase_margin)                                      |
| Phase Margin Frequency                      | meas(phase_margin_freq)                                 |
| Unity Gain Frequency                        | meas(unity_gain_freq)                                   |
| Loop Gain Min Frequency                     | meas(loop_gain_minifreq)                                |